

EXPLORATORY DATA ANALYSIS

Where Should We Invest in Wind Energy?

A data-driven look at U.S. wind turbine performance, by region and design



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BACKGROUND

Wind Energy Is Booming — But Location Is Everything

- Wind is one of the fastest-growing sources of U.S. electricity, with tens of thousands of turbines installed nationwide
- Investment returns depend heavily on where a farm is built — turbine size, local wind strength, and electricity prices all matter
- This analysis covers turbines across all four U.S. Census Regions: Northeast, Midwest, South, and West
- The goal: pinpoint the region and turbine design that gives investors the strongest, most reliable energy output



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U.S. Census Regions Analyzed

Northeast · Midwest · South · West

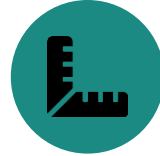
PROBLEM STATEMENT

The Business Question We Set Out to Answer

A private-equity firm wants to acquire U.S. wind farms, but needs to know which regions and turbine designs actually drive the strongest energy output.



Which region has the most capacity and the highest output?



How do turbine specs — rotor size, hub height — affect energy output?



Is there a trade-off between wind strength and electricity prices?



Objective: Recommend the strongest region and turbine profile for the next acquisition.

How We Built the Analysis

70,808

Turbine records in the raw U.S. Wind Turbine Database

64,562

Turbines remaining after cleaning — our final analysis set

4

External datasets merged in: state wind speed, electricity rates & bills

Methodology

1

Mapped each turbine's state to its U.S. Census Region

2

Merged in state-level wind speed and electricity rate/bill data

3

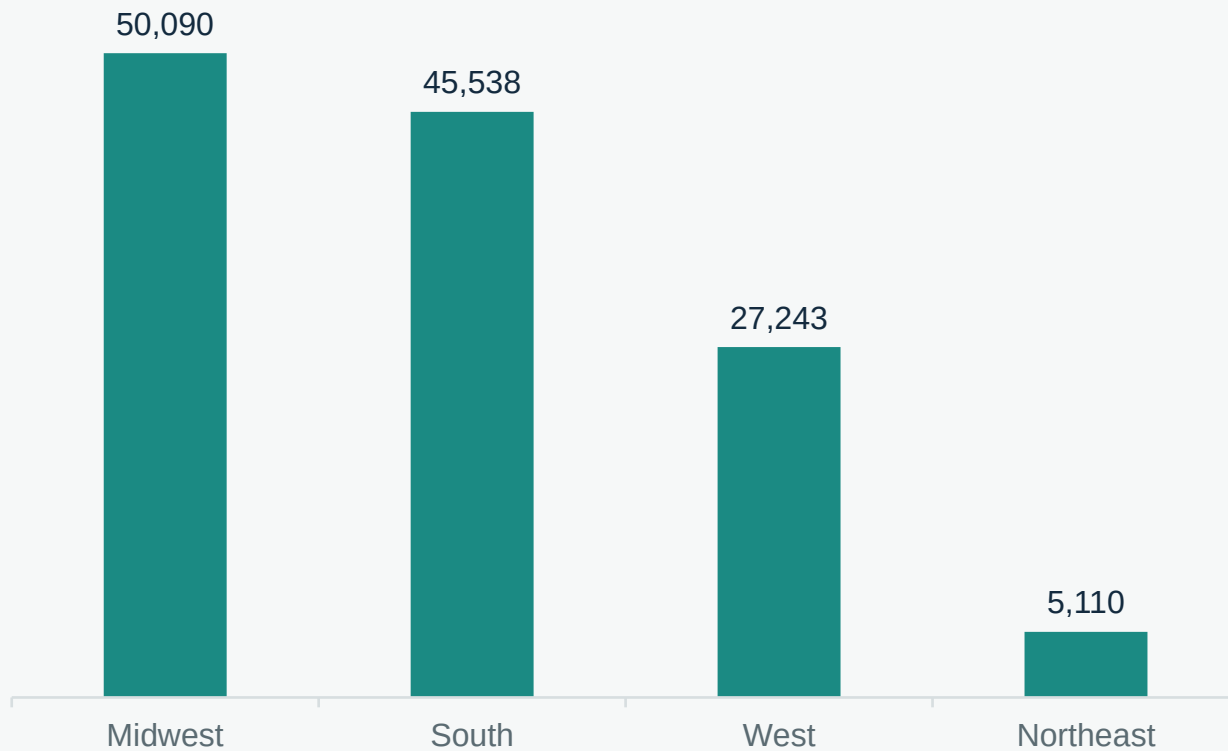
Standardized manufacturer names and removed incomplete specs

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Engineered a Net Generation estimate (capacity × estimated efficiency) as a proxy for real production data

KEY FINDING #1

Installed Capacity Is Concentrated in the Midwest & South

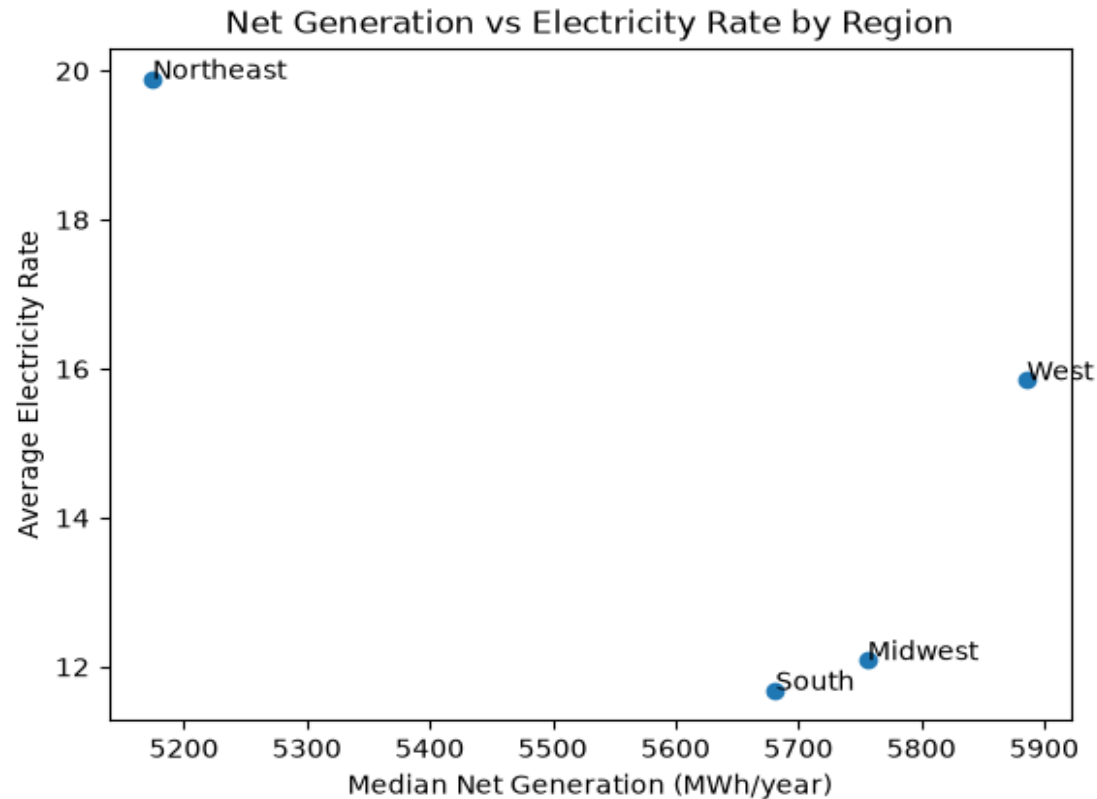


- Midwest leads with ~50,100 MW of installed capacity — nearly 10x the Northeast
- South is close behind at ~45,500 MW; West follows at ~27,400 MW
- Northeast trails far behind at just ~5,100 MW

Takeaway: Most of the industry's scale already sits in the Midwest and South.

KEY FINDING #2

Strong Output Doesn't Always Mean Cheap Electricity

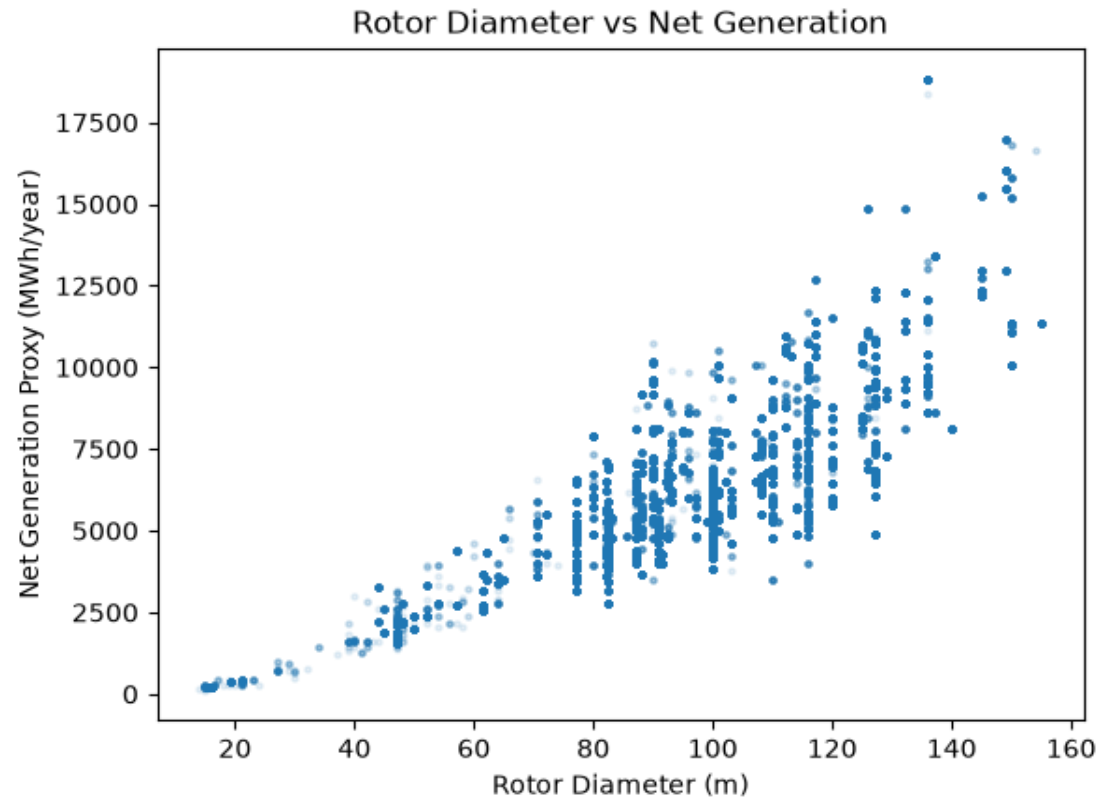


- Northeast: lowest energy output AND the highest electricity rate — weakest region on both fronts
- Midwest & South: strong output paired with the cheapest electricity — the best combination
- West: highest output per turbine, but electricity rates run higher too

Takeaway: The best regions balance output with electricity cost — not wind speed alone.

KEY FINDING #3

Bigger Rotor Blades Are the Clearest Driver of Output



- Turbines with larger rotor blades consistently generate more energy
- This relationship holds true almost regardless of region
- Larger rotors capture more wind — a larger swept area converts to more power

Takeaway: Rotor diameter is the strongest single technical lever for higher output we found in the data.

CONCLUSIONS

What the Data Tells Us



Scale sits in the Midwest & South, but per-turbine efficiency is close across all four regions (~5,100-5,900 MWh/yr)



Midwest & South offer the best balance of strong output and low electricity rates



West generates the most per turbine — but at a noticeably higher electricity price

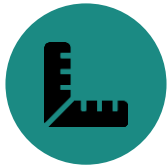


Rotor diameter is the clearest technical lever for higher energy output

Bottom line: The Northeast is the weakest performer overall — lowest output, highest cost, fewest turbines.

RECOMMENDATIONS

Where We Recommend Focusing Next



Prioritize rotor diameters of 100m+

This single design factor is the strongest link to higher output, regardless of region



Focus acquisitions on the Midwest & South

Best combination of strong generation and the lowest electricity rates



Consider the West selectively

Highest output per turbine, but only if a premium electricity price is acceptable

Deprioritize: The Northeast — weakest wind, highest electricity cost, fewest turbines.

Next steps: Drill into state-level data for precise site targeting, and validate against real production data once available.